

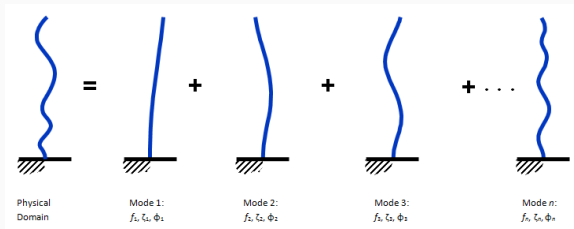
Eigenvalue problems of vibrations and stability

Analysis of free and forced vibrations

ME473 Dynamic finite element analysis of structures

Stefano Burzio

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Where do we stand?

Week	Module	Lecture topic	Mini-projects
1	Linear elastodynamics	Strong and weak forms	
2		Galerkin and FE methods	
3		Systematisation of the procedure	Groups formation
4		3d elements, numerical integration	Project 1 statement
5	Special structural elements	Bars and trusses	
6		Beams, frames and grids	
7		Kirchhoff-Love plates	
8		Reissner-Mindlin plates	
9		Shells	Project 1 submission
10	Free and forced vibrations	Analysis of free vibrations	Project 2 statement

Summary

- Recap week 9
- Modal properties of conservative systems
- Example: Rayleigh's quotient for axial vibrations of a free-free bar
- Numerical modal extraction algorithms for conservative systems

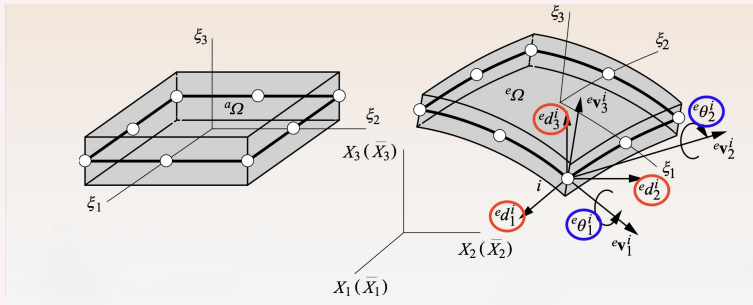
Recommended readings

- (N) Neto et al., Engineering Computation of Structures (chap. 2.5)
- (P) Petyt, Introduction to finite element vibration analysis (chap. 11)
- (G) Gmür, Dynamique des structures (§4.1 and §4.2)

Recap week 9

Vibrations of shells

Shell element

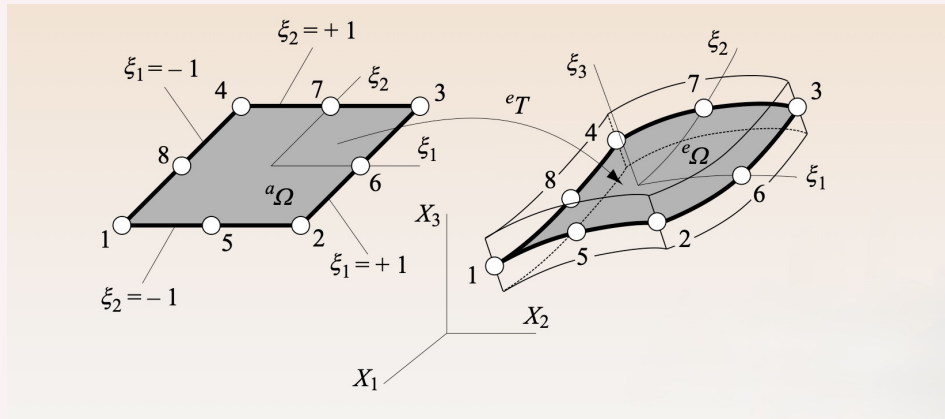


- have 5 DOFs per node, no rotation $^e \theta_3^i$.
- lead to huge computational time savings since allow modeling with fewer mesh elements.
- less prone to negative Jacobian errors which might occur when using extremely thin 3d solid elements.

Shape functions matrix

$$\begin{aligned}
 \mathbf{u}^h(\boldsymbol{\xi}) &= \sum_{i=1}^{e_p} {}^a h_i(\xi_1, \xi_2) \left[{}^e \mathbf{d}^i + \frac{1}{2} \xi_3 {}^e t_3^i \left(-{}^e \theta_1^i {}^e \mathbf{v}_2^i + {}^e \theta_2^i {}^e \mathbf{v}_1^i \right) \right] \\
 &= \sum_{i=1}^{e_p} {}^a \mathbf{H}_i(\boldsymbol{\xi}) {}^e \mathbf{q}^i(t) \\
 &= \sum_{i=1}^{e_p} \underbrace{\begin{bmatrix} {}^a h_i & 0 & 0 & -\frac{1}{2} \xi_3 {}^e t_3^i {}^a h_i {}^e v_{21}^i & \frac{1}{2} \xi_3 {}^e t_3^i {}^a h_i {}^e v_{11}^i \\ 0 & {}^a h_i & 0 & -\frac{1}{2} \xi_3 {}^e t_3^i {}^a h_i {}^e v_{22}^i & \frac{1}{2} \xi_3 {}^e t_3^i {}^a h_i {}^e v_{12}^i \\ 0 & 0 & {}^a h_i & -\frac{1}{2} \xi_3 {}^e t_3^i {}^a h_i {}^e v_{23}^i & \frac{1}{2} \xi_3 {}^e t_3^i {}^a h_i {}^e v_{13}^i \end{bmatrix}}_{{}^a \mathbf{H}_i} \underbrace{\begin{bmatrix} {}^e d_1^i \\ {}^e d_2^i \\ {}^e d_3^i \\ {}^e \theta_1^i \\ {}^e \theta_2^i \end{bmatrix}}_{{}^e \mathbf{q}^i}
 \end{aligned}$$

Example: 8 nodes quadrangular shell element

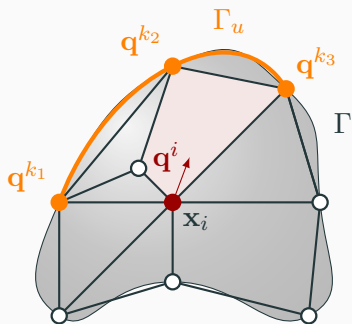


Analysis of free vibrations

Free vibrations of non-rotating conservative systems

The discretization of linear three-dimensional elastodynamics, as well as the analysis of vibrations in beams, plates and shells, all lead to a system of ODE:

$$\mathbf{M}\ddot{\mathbf{q}}(t) + \mathbf{K}\mathbf{q}(t) = \mathbf{r}(t),$$



Free vibration: no external forcing is applied, i.e. $\mathbf{r}(t) = \mathbf{0}$.

- Generalized nodal displacements:

$$\mathbf{q}(t) = [\mathbf{q}^1(t), \dots, \mathbf{q}^n(t)]^T.$$

- **Boundary conditions:** $\mathbf{q}^k = \hat{\mathbf{q}}^k$ for all k such that $\mathbf{x}_k \in \Gamma_u$.
- **Initial conditions:** $\mathbf{q}(0) = \mathbf{u}_0$ and $\dot{\mathbf{q}}(0) = \mathbf{v}_0$

Harmonic response

The solution for a free and undamped discrete vibration problem

$$\mathbf{M}\ddot{\mathbf{q}}(t) + \mathbf{K}\mathbf{q}(t) = \mathbf{0}$$

is sinusoidal, described via an harmonic function:

$$\mathbf{q}(t) = \alpha \mathbf{p} \cos(\omega t + \varphi)$$

- $\mathbf{p}$: mode shape (modal vector)
- φ : phase
- ω : natural frequency
- α : scaling factor
- **Solutions are defined up to a scalar factor: α .**
- Three scalars unknowns α , ω , φ and one unknown vector $\mathbf{p}$.

Generalized eigenvalue problem

Substituting the proposed ansatz into the equation of motion yields a generalized eigenvalue problem:

$$\begin{aligned}\alpha(\mathbf{K} - \omega^2\mathbf{M})\mathbf{p} \cos(\omega t - \varphi) &= 0 \\ (\mathbf{K} - \omega^2\mathbf{M})\mathbf{p} &= 0\end{aligned}$$

Solving the eigenvalue problem:

- *Eigenvalue:* $\lambda_j = \omega_j^2$ are the roots of the characteristic polynomial:

$$\det(\mathbf{K} - \omega^2\mathbf{M}) = 0.$$

- *Eigenvector:* $\mathbf{p}_j$ are the solution of the equation

$$(\mathbf{K} - \lambda_j\mathbf{M})\mathbf{p}_j = \mathbf{0}.$$

General solution via modal superposition

- Once the set of eigenvalue-eigenvector pairs $(\omega_j^2, \mathbf{p}_j)$ have been determined, the linearity of the system allows the general solution to be expressed as a superposition of modal contributions:

$$\mathbf{q}(t) = \sum_{j=1}^n \alpha_j \mathbf{p}_j \cos(\omega_j t + \varphi_j)$$

- The constants α_j and φ_j are determined from the initial conditions: $\mathbf{q}(0) = \mathbf{u}_0$ and $\dot{\mathbf{q}}(0) = \mathbf{v}_0$ leading to the system:

$$\begin{aligned} \sum_{j=1}^n \alpha_j \mathbf{p}_j \cos(\varphi_j) &= \mathbf{u}_0 \\ \sum_{j=1}^n \alpha_j \omega_j \mathbf{p}_j \sin(\varphi_j) &= -\mathbf{v}_0 \end{aligned}$$

Fundamental properties of generalized eigenvalue problem

- If both $\mathbf{K}$ and $\mathbf{M}$ are symmetric and strictly positive definite, then all eigenvalues λ_j of the generalized eigenvalue problem

$$\mathbf{K}\mathbf{p}_j = \lambda_j \mathbf{M}\mathbf{p}_j$$

are real and positive.

- Thus we can define the angular frequencies

$$\omega_j = \sqrt{\lambda_j}$$

and

$$0 < \omega_1 \leq \omega_2 \leq \omega_3 \leq \cdots \leq \omega_n.$$

Rigid body modes

In the semi-discrete weak form obtained via finite element discretization:

- The **mass matrix** $\mathbf{M}$ is symmetric and strictly positive definite.
- The **stiffness matrix** $\mathbf{K}$ is symmetric and positive semi-definite:

$$\mathbf{K}\mathbf{p} = \mathbf{0} \quad \text{for certain nonzero vectors } \mathbf{p}.$$

Consequently, the eigenvalues ω_j^2 of the generalized eigenvalue problem are all real and non-negative:

$$0 \leq \omega_1 \leq \omega_2 \leq \cdots \leq \omega_n.$$

Rigid body modes: zero eigenvalues (i.e., $\omega_j = 0$) correspond to *rigid body motions*, where the system undergoes displacement without internal deformation.

Spectral offset technique

To ensure that the stiffness matrix is *strictly* positive definite, a **spectral shift** strategy is employed. Given $\sigma > 0$, define the modified stiffness matrix as:

$$\mathbf{K}_\sigma = \mathbf{K} + \sigma \mathbf{M}$$

- Consider the offset eigenvalue problem: $(\mathbf{K}_\sigma - \lambda_\sigma \mathbf{M})\mathbf{p} = \mathbf{0}$
- Substituting the definition of $\mathbf{K}_\sigma$:

$$(\mathbf{K} + \sigma \mathbf{M} - \lambda_\sigma \mathbf{M})\mathbf{p} = (\mathbf{K} + (\sigma - \lambda_\sigma) \mathbf{M})\mathbf{p} = \mathbf{0}$$

- Comparing with $(\mathbf{K} - \lambda \mathbf{M})\mathbf{p} = \mathbf{0}$, we identify: $\lambda_\sigma = \lambda + \sigma > 0$.

Eigenvalues are shifted by σ , but the eigenvectors remain the same.

$$\textit{Empirical rule:} \quad \sigma \approx \frac{1}{100} \frac{\text{tr}(\mathbf{K})}{\text{tr}(\mathbf{M})}$$

Orthogonality of mode shapes

Let $\mathbf{p}_i$ and $\mathbf{p}_j$ two eigenvectors corresponding to two *distinct* eigenvalues λ_i and λ_j , then

$$\mathbf{p}_i^T \mathbf{M} \mathbf{p}_j = 0 \quad \text{and} \quad \mathbf{p}_i^T \mathbf{K} \mathbf{p}_j = 0 \quad (i \neq j).$$

Consequence: two different harmonic responses:

$$\mathbf{q}_i(t) = \alpha_i \mathbf{p}_i \cos(\omega_i t + \varphi_i) \quad \text{and} \quad \mathbf{q}_j(t) = \alpha_j \mathbf{p}_j \cos(\omega_j t + \varphi_j)$$

are $\mathbf{M}$ - and $\mathbf{K}$ -orthogonal:

$$\mathbf{q}_i^T \mathbf{M} \mathbf{q}_j = 0 \quad \text{and} \quad \mathbf{q}_i^T \mathbf{K} \mathbf{q}_j = 0.$$

The virtual work of inertial and elastic forces of a given mode, along the displacement given by a different mode, is zero.

Normalization of mode shapes

Let $\mathbf{p}_i$ and $\mathbf{p}_j$ two eigenvectors corresponding to the eigenvalues λ_i and λ_j , then

$$\mathbf{p}_i^T \mathbf{M} \mathbf{p}_j = \mu_i \delta_{ij} \quad \text{and} \quad \mathbf{p}_i^T \mathbf{K} \mathbf{p}_j = \kappa_i \delta_{ij}$$

where

- μ_i = modal mass of mode i ,
- κ_i = modal stiffness of mode i ,
- δ_{ij} = Kronecker delta.

Usual normalization choice: orthonormalization

$$\mu_i = 1 \quad \Rightarrow \quad \lambda_i = \frac{\kappa_i}{\mu_i} = \omega_i^2.$$

Orthonormalization of mode shapes

Let $\mathbf{p}_i$ and $\mathbf{p}_j$ two eigenvectors corresponding to the eigenvalues λ_i and λ_j , then

$$\mathbf{p}_i^T \mathbf{M} \mathbf{p}_j = \delta_{ij} \quad \text{and} \quad \mathbf{p}_i^T \mathbf{K} \mathbf{p}_j = \omega_i^2 \delta_{ij}$$

where δ_{ij} represent Kronecker symbol.

We store the modal vectors $\mathbf{p}_i$ in a so-called *modal matrix* $\mathbf{P}$:

$$\mathbf{P} = \left[\begin{array}{c|c|c|c} \mathbf{p}_1 & \mathbf{p}_2 & \dots & \mathbf{p}_n \end{array} \right]$$

then

$$\mathbf{P}^T \mathbf{M} \mathbf{P} = \mathbf{I} \quad \text{and} \quad \mathbf{P}^T \mathbf{K} \mathbf{P} = \mathbf{\Lambda}$$

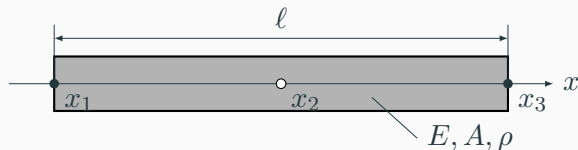
where $\mathbf{I}$ is the identity matrix of order n and $\mathbf{\Lambda}$ the spectral matrix:

$$\mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_n) = \text{diag}(\omega_1^2, \dots, \omega_n^2).$$

Example: longitudinal vibrations of free-free bar

Longitudinal vibrations of free-free bar

Consider a free-free bar discretized by one bilinear element.



- ℓ length
- A cross-sectional area
- E Young's modulus
- ρ material density

Objectives:

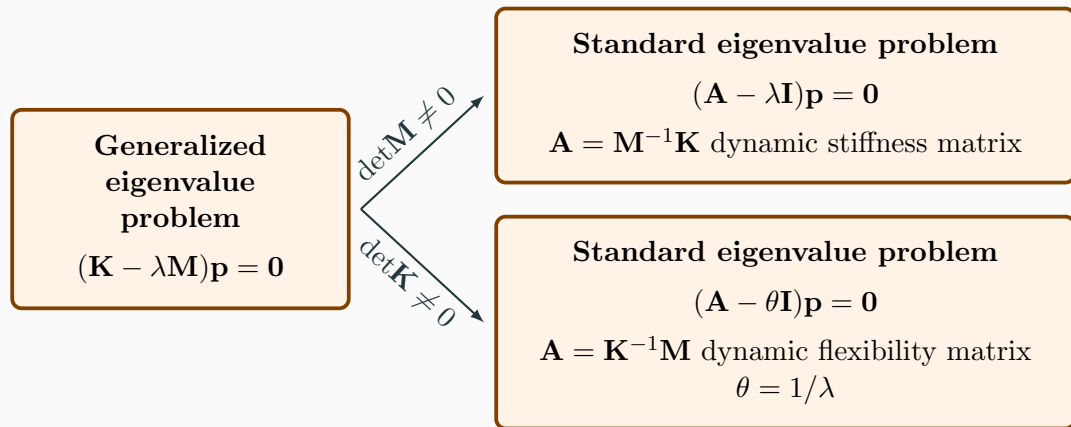
- ① compute the approximated natural frequencies and corresponding mode shapes,
- ② verify the presence of a rigid body mode,
- ③ check the orthonormality of the mode shapes.

MATLAB example - longitudinal vibrations of free-free bar

► [Go to Matlab Drive](#)

Eigenproblem solution methods

Transformation to standard form by inversion



The matrices $\mathbf{M}^{-1}\mathbf{K}$ and $\mathbf{K}^{-1}\mathbf{M}$ are not symmetric.

Transformation to standard form by decomposition

To reduce to a standard eigenvalue problem:

- Let $\mathbf{L}$ be the *Cholesky factor* of $\mathbf{M}$: $\mathbf{M} = \mathbf{L}\mathbf{L}^T$.

Since $\mathbf{M}$ is symmetric positive-definite, $\mathbf{L}$ is a real lower triangular matrix with positive diagonal entries.

- Define the change of variables $\mathbf{v} = \mathbf{L}^T \mathbf{p}$. Substituting into the original problem and multiplying by $\mathbf{L}^{-1}$ yields:

$$(\mathbf{K} - \omega^2 \mathbf{L}\mathbf{L}^T) \mathbf{p} = 0 \quad \Rightarrow \quad (\mathbf{L}^{-1} \mathbf{K} \mathbf{L}^{-T} - \omega^2 \mathbf{I}) \mathbf{v} = 0.$$

- **Result:** standard eigenvalue problem for symmetric and positive definite matrix $\mathbf{A} = \mathbf{L}^{-1} \mathbf{K} \mathbf{L}^{-T}$:

$$(\mathbf{A} - \lambda \mathbf{I}) \mathbf{v} = \mathbf{0}.$$

Transformation to standard form by decomposition

- Cholesky decomposition can be applied to $\mathbf{K}$ as well, since $\mathbf{K}$ is symmetric positive-definite:

$$\mathbf{K} = \mathbf{L}\mathbf{L}^T.$$

- **Result:** standard eigenvalue problem for symmetric and positive definite matrix $\mathbf{A} = \mathbf{L}^{-1}\mathbf{M}\mathbf{L}^{-T}$, eigenvalues $\theta = 1/\lambda$ and eigenvectors $\mathbf{v} = \mathbf{L}^T\mathbf{p}$:

$$(\mathbf{A} - \theta\mathbf{I})\mathbf{v} = \mathbf{0}.$$

Small and large scale problems

Small-scale problems

- system matrices are of modest size $n \leq 250$ or $250 \leq n \leq 2500$ and small bandwidth matrices.
- An explicit reduction to standard eigenvalue form is typically employed.

Large-scale problems

- $n \geq 2500$.
- The most effective algorithms are subspace iteration, Lanczos' method, Guyan-Irons method, Arnoldi method and Davidson method.

Numerical modal extraction algorithms

1. Rayleigh's quotient

Definition and first properties

Let $\mathbf{w}$ be a vector

$$\mathcal{R}(\mathbf{w}) = \frac{\mathbf{w}^T \mathbf{K} \mathbf{w}}{\mathbf{w}^T \mathbf{M} \mathbf{w}}$$

Properties

- ① Homogeneity. Let α a non-zero constant, then

$$\mathcal{R}(\alpha \mathbf{w}) = \frac{(\alpha \mathbf{w})^T \mathbf{K} (\alpha \mathbf{w})}{(\alpha \mathbf{w})^T \mathbf{M} (\alpha \mathbf{w})} = \frac{\alpha^2 \mathbf{w}^T \mathbf{K} \mathbf{w}}{\alpha^2 \mathbf{w}^T \mathbf{M} \mathbf{w}} = \mathcal{R}(\mathbf{w})$$

- ② Rayleigh quotient of an eigenvector:

$$\mathcal{R}(\mathbf{w} = \mathbf{p}_i) = \frac{\mathbf{p}_i^T \mathbf{K} \mathbf{p}_i}{\mathbf{p}_i^T \mathbf{M} \mathbf{p}_i} = \lambda_i = \omega_i^2$$

Stationarity of Rayleigh's quotient

First variation of Rayleigh's quotient:

$$\begin{aligned}\delta\mathcal{R}(\mathbf{w}) &= \frac{2(\delta\mathbf{w}^T\mathbf{K}\mathbf{w})(\mathbf{w}^T\mathbf{M}\mathbf{w}) - 2(\mathbf{w}^T\mathbf{K}\mathbf{w})(\delta\mathbf{w}^T\mathbf{M}\mathbf{w})}{(\mathbf{w}^T\mathbf{M}\mathbf{w})^2} \\ &= \frac{2\delta\mathbf{w}^T}{\mathbf{w}^T\mathbf{M}\mathbf{w}} \left(\mathbf{K}\mathbf{w} - \frac{\mathbf{w}^T\mathbf{K}\mathbf{w}}{\mathbf{w}^T\mathbf{M}\mathbf{w}}\mathbf{M}\mathbf{w} \right) \\ &= \frac{2\delta\mathbf{w}^T}{\mathbf{w}^T\mathbf{M}\mathbf{w}} (\mathbf{K} - \mathcal{R}(\mathbf{w})\mathbf{M})\mathbf{w}\end{aligned}$$

If $\mathbf{w} = \mathbf{p}_i$ then

$$(\mathbf{K} - \mathcal{R}(\mathbf{w})\mathbf{M})\mathbf{w} = (\mathbf{K} - \lambda_i\mathbf{M})\mathbf{p}_i = 0$$

Stationary Rayleigh's quotient in the vicinity of an modal shape !

Rayleigh principle

Modal expansion:

$$\mathbf{w} = z_1 \mathbf{p}_1 + z_2 \mathbf{p}_2 + \cdots + z_i \mathbf{p}_i + \cdots + z_n \mathbf{p}_n = \mathbf{P} \mathbf{z}$$

where

$$\mathbf{P} = [\mathbf{p}_1, \mathbf{p}_2, \dots, \mathbf{p}_i, \dots, \mathbf{p}_n] \quad \text{modal matrix}$$

$$\mathbf{z} = [z_1, z_2, \dots, z_i, \dots, z_n]^T \quad \text{coefficient vector}$$

Then

$$\mathcal{R}(\mathbf{w} = \mathbf{P} \mathbf{z}) = \frac{\mathbf{z}^T \mathbf{P}^T \mathbf{K} \mathbf{P} \mathbf{z}}{\mathbf{z}^T \mathbf{P}^T \mathbf{M} \mathbf{P} \mathbf{z}} = \frac{\mathbf{z}^T \boldsymbol{\Lambda} \mathbf{z}}{\mathbf{z}^T \mathbf{z}} = \frac{\sum_{i=1}^n \lambda_i z_i^2}{\sum_{i=1}^n z_i^2}$$

If, without loss of generality, we impose $\mathbf{w}^T \mathbf{M} \mathbf{w} = 1$, then $\mathbf{z}^T \mathbf{z} = \sum_{i=1}^n z_i^2 = 1$ and

$$\mathcal{R}(\mathbf{w} = \mathbf{P} \mathbf{z}) = \sum_{i=1}^n \lambda_i z_i^2$$

Rayleigh principle (continued)

$$\begin{aligned}\mathcal{R}(\mathbf{w}) &= \sum_{i=1}^n \lambda_i z_i^2 = \lambda_j - \lambda_j + \sum_{i=1}^n \lambda_i z_i^2 = \lambda_j - \lambda_j \left(\sum_{i=1}^n z_i^2 \right) + \sum_{i=1}^n \lambda_i z_i^2 \\ &= \lambda_j + \sum_{i=1}^n (\lambda_i - \lambda_j) z_i^2\end{aligned}$$

With eigenvalues ordered as $\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_i \leq \dots \leq \lambda_n$, we have:

- If $j = 1$, then $\lambda_i - \lambda_1 \geq 0$ for all i , hence

$$\mathcal{R}(\mathbf{w}) = \lambda_1 + \sum_{i=1}^n (\lambda_i - \lambda_1) z_i^2 \geq \lambda_1.$$

- If $j = n$, then $\lambda_i - \lambda_n \leq 0$ for all i , hence

$$\mathcal{R}(\mathbf{w}) = \lambda_n + \sum_{i=1}^n (\lambda_i - \lambda_n) z_i^2 \leq \lambda_n.$$

Rayleigh quotient bounding theorem

- Courant-Fischer formulas:

$$\mathcal{R}(\mathbf{w}) \geq \lambda_1 = \min_{\mathbf{w}} \mathcal{R}(\mathbf{w}) \quad \mathcal{R}(\mathbf{w}) \leq \lambda_n = \max_{\mathbf{w}} \mathcal{R}(\mathbf{w})$$

- Bounding theorem:

$$\lambda_1 \leq \mathcal{R}(\mathbf{w}) \leq \lambda_n$$

Modal frequencies and shapes search via Rayleigh quotient minimization

(1) Find λ_1 by minimization:

$$\lambda_1 = \min_{\mathbf{w}} \mathcal{R}(\mathbf{w})$$

(2) Find $\mathbf{p}_1$ by solving $(\mathbf{K} - \lambda_1 \mathbf{M})\mathbf{p}_1 = 0$.

(3) Find λ_2 by minimization:

$$\lambda_2 = \min_{\mathbf{w}} [\mathcal{R}(\mathbf{w}); \mathbf{w}^T \mathbf{M} \mathbf{p}_1 = 0]$$

(4) Find $\mathbf{p}_2$ by solving $(\mathbf{K} - \lambda_2 \mathbf{M})\mathbf{p}_2 = 0$.

$\vdots$

(j) Find λ_j by minimization:

$$\lambda_j = \min_{\mathbf{w}} [\mathcal{R}(\mathbf{w}); \mathbf{w}^T \mathbf{M} \mathbf{p}_i = 0, i = 1, 2, \dots, j-1]$$

(j+1) Find $\mathbf{p}_j$ by solving $(\mathbf{K} - \lambda_j \mathbf{M})\mathbf{p}_j = 0$.

Example: Rayleigh's quotient for
longitudinal vibrations of a free-free bar

MATLAB example - longitudinal vibrations of free-free bar

► [Go to Matlab Drive](#)

Numerical modal extraction algorithms

2. Subspace iteration

Inverse iteration algorithm

Goal: compute the eigenvector associated with the smallest eigenvalue of the generalized eigenproblem $\mathbf{K}\mathbf{p} = \lambda\mathbf{M}\mathbf{p}$

Inputs:

- $\mathbf{K}$, $\mathbf{M}$: stiffness and mass matrices
- σ : spectral shift (optional)
- $p^{(0)}$: initial guess vector (non-zero)
- ε : convergence tolerance

Algorithm:

- ① If $\mathbf{K}$ is singular, use shift: set $\mathbf{K}_\sigma = \mathbf{K} + \sigma\mathbf{M}$
- ② For $k = 1, 2, \dots$ until convergence:
 - Solve $\mathbf{K}_\sigma p^{(k)} = \mathbf{M} p^{(k-1)}$
 - Normalize: $p^{(k)} \leftarrow p^{(k)} / \|p^{(k)}\|$
 - Check convergence: $\|p^{(k)} - p^{(k-1)}\| < \varepsilon$

Output: approximated eigenvector $p^{(k)}$

Subspace iteration method

Goal: compute the first $m \ll n$ eigenpairs $(\mathbf{p}_i, \lambda_i)$ of the generalized eigenproblem.

Inputs:

- $\mathbf{K}, \mathbf{M}$: stiffness and mass matrices
- $\mathbf{P}^{(0)} \in \mathbb{R}^{n \times q}$: initial guess (matrix with $q > m$ linearly independent vectors)
- σ : spectral shift (optional)
- ε : convergence tolerance

Output:

- Approximated eigenvectors: $\mathbf{P}^{(k)} = [\mathbf{p}_1^{(k)}, \dots, \mathbf{p}_q^{(k)}]$
- Approximated eigenvalues: $\mathbf{\Lambda}^{(k)} = \text{diag}(\lambda_1^{(k)}, \dots, \lambda_q^{(k)})$

Algorithm:

- ① If $\mathbf{K}$ is singular, use shift: set $\mathbf{K}_\sigma = \mathbf{K} + \sigma\mathbf{M}$
- ② For $k = 1, 2, \dots$ until convergence:
 - Do steps 1, 2a, 2b and 2c
 - Check convergence

Subspace iteration steps

- ① **Step 1:** Simultaneous inverse iteration on $q > m$ vectors: find the $(n \times q)$ matrix $\overline{\mathbf{P}}^{(k)}$ such that

$$\mathbf{K}\overline{\mathbf{P}}^{(k)} = \mathbf{M}\mathbf{P}^{(k-1)}$$

- ② **Step 2a:** Compute projected stiffness and mass matrices:

$$\mathbf{K}^{(k)} = (\overline{\mathbf{P}}^{(k)})^T \mathbf{K} \overline{\mathbf{P}}^{(k)}, \quad \mathbf{M}^{(k)} = (\overline{\mathbf{P}}^{(k)})^T \mathbf{M} \overline{\mathbf{P}}^{(k)}$$

- ③ **Step 2b:** Solve $(q \times q)$ generalized eigenvalue problem: Find the modal matrix and the spectral matrix such that

$$\mathbf{K}^{(k)} \mathbf{Z}^{(k)} = \mathbf{M}^{(k)} \mathbf{Z}^{(k)} \mathbf{\Lambda}^{(k)}$$

- ④ **Step 2c:** Orthogonalization:

$$\mathbf{P}^{(k)} = \overline{\mathbf{P}}^{(k)} \mathbf{Z}^{(k)}$$

Subspace algorithm - step 1

Suppose that the Step 1 is replaced by a simultaneous inverse iteration on m eigenvectors:

$$\mathbf{P}^{(k)} = (\mathbf{K}^{-1}\mathbf{M})\mathbf{P}^{(k-1)} = \dots = (\mathbf{K}^{-1}\mathbf{M})^k \mathbf{P}_0$$

Define the subspace $\mathcal{S}^{(k)}$ of rank q , spanned by the vectors $\{\mathbf{p}_i^{(k)}\}$.

$\mathbf{P}^{(k)} = [\mathbf{p}_1^{(k)}, \dots, \mathbf{p}_q^{(k)}]$ forms a *non-orthogonal* basis of $\mathcal{S}^{(k)}$.

- ✗ All columns of $\mathbf{P}^{(k)}$ tend toward $\mathbf{p}_1$
- ✗ Collinearity if no orthogonalization is applied !

- **Orthogonalization** of vectors $\mathbf{p}_i^{(k)}$ at each iteration
- Use, for instance, Gram-Schmidt method (*Note: this step is computationally expensive*)

Subspace algorithm - step 2a

- Orthogonalization by minimization of the Rayleigh quotient:

$$\mathcal{R}(\mathbf{w}^{(k)}) = \frac{(\mathbf{w}^{(k)})^T \mathbf{K} \mathbf{w}^{(k)}}{(\mathbf{w}^{(k)})^T \mathbf{M} \mathbf{w}^{(k)}}$$

- Let $\mathbf{w}^{(k)} = \overline{\mathbf{P}^{(k)}} \mathbf{z}^{(k)}$

- Projected Rayleigh's quotient:

$$\mathcal{R}(\mathbf{w}^{(k)}) = \frac{(\mathbf{z}^{(k)})^T \mathbf{K}^{(k)} \mathbf{z}^{(k)}}{(\mathbf{z}^{(k)})^T \mathbf{M}^{(k)} \mathbf{z}^{(k)}}$$

where

$$\mathbf{K}^{(k)} = (\overline{\mathbf{P}^{(k)}})^T \mathbf{K} \overline{\mathbf{P}^{(k)}}, \quad \mathbf{M}^{(k)} = (\overline{\mathbf{P}^{(k)}})^T \mathbf{M} \overline{\mathbf{P}^{(k)}}$$

Subspace algorithm - step 2b

- Minimization of the Projected Rayleigh's quotient (generalized eigenvalue problem of dimension $q \times q$)
- Stationary condition:

$$\delta \mathcal{R}(\mathbf{w}^{(k)}) = 0 \quad \Rightarrow \quad \mathbf{K}^{(k)} \mathbf{z}^{(k)} = \lambda^{(k)} \mathbf{M}^{(k)} \mathbf{z}^{(k)}$$

- Solve via transformation method (e.g., Jacobi method):

$$\mathbf{K}^{(k)} \mathbf{Z}^{(k)} = \mathbf{M}^{(k)} \mathbf{Z}^{(k)} \mathbf{\Lambda}^{(k)}$$

- Ritz vectors and values:

$$\mathbf{Z}^{(k)} = [\mathbf{z}_1^{(k)}, \dots, \mathbf{z}_q^{(k)}], \quad \text{and} \quad \mathbf{\Lambda}^{(k)} = \text{diag}(\lambda_1^{(k)}, \dots, \lambda_q^{(k)})$$

Subspace algorithm - step 2c

- Update the modal matrix:

$$\mathbf{P}^{(k)} = \overline{\mathbf{P}^{(k)}} \mathbf{Z}^{(k)}$$

- Orthogonality check:

$$(\mathbf{P}^{(k)})^T \mathbf{M} \mathbf{P}^{(k)} = \mathbf{I}$$